

XING HAN

[Homepage] ◊ [Google Scholar] ◊ [LinkedIn] ◊ [xhan56@jh.edu] ◊ +1 (512) 987 5026 ◊ Mountain View, CA, USA

RESEARCH INTERESTS

I build multimodal foundation models and agents that acquire the core competencies of human experts: integrating heterogeneous evidence as it unfolds over time, reasoning about “what-if” trajectories, and staying reliable as the situation evolves. My work bridges core machine learning methodology with high-stakes applications such as medicine.

- **Multimodal Foundation Models:** multimodal fusion, cross-modal interactions, clinical foundation models
- **Counterfactual Reasoning & Simulation:** counterfactual simulation, physiology-constrained generative models
- **Trustworthy & Enduring Deployment:** uncertainty quantification, sequential monitoring, continual learning
- **Foundations of Machine Learning:** neural scaling laws, time-series sequence models, interpretability

EDUCATION

M.Sc - Ph.D. in Electrical and Computer Engineering

The University of Texas at Austin

August 2023

Austin, TX, USA

Advisor: Joydeep Ghosh Committee: Aryan Mokhtari, Atlas Wang, Mingyuan Zhou, Nhat Ho

Dissertation: Reliable Predictions for Structured and Corrupted Data

B.Eng. in Electrical and Electronics Engineering

The University of Edinburgh

July 2017

Edinburgh, UK

Advisor: Jiabin Jia GPA: 3.92 (graduated with first-class honor)

University of Edinburgh “2+2” Scholarship in 2015 & 2016

PUBLICATIONS

XH denotes the author; * denotes equal contribution; ✉ denotes corresponding author.

Selected Papers

[1] **On the Invariance and Generality of Neural Scaling Laws**

Top ~3% NeurIPS submissions by reviewer rating

XH, Z. Liu, S. Saria, and P. Liang

Preprint

2026

[2] **FLAME: Adaptive Mixture-of-Experts for Continual Multimodal Multi-Task Learning**

XH, S. Chaudhari, T. Ranade, R. Chellappa, and S. Saria

Preprint

2026

[3] **Massively Multimodal Foundation Models: A Frameworks for Capturing Interactions with Specialized Mixture-of-Experts**

Top ~10% of accepted papers by reviewer rating

XH, H. Chung, J. Ghosh, P. Liang, and S. Saria

International Conference on Learning Representations (ICLR)

2026

[4] **QoQ-Med3: Robust Multimodal Clinical Analysis Foundation Model with Reasoning**

W. Dai, J. She, J. Cheong, **XH**, C. Harris, H. Wei, F. Vahedifard, S. Saria, R. Stevens, and P. Liang

NPJ Digital Medicine

2026

- [5] **WATCH: Adaptive Monitoring for AI Deployments via Weighted-Conformal Martingales**
D. Prinster [✉], **XH** [✉], A. Liu, and S. Saria
International Conference on Machine Learning (ICML) 2025
- [6] **FuseMoE: Mixture-of-Experts Transformers for Fleximodal Fusion**
Over 149 citations as of July 2026
XH, H. Nguyen, C. Harris, N. Ho, and S. Saria
Neural Information Processing Systems (NeurIPS) 2024
- [7] **Designing Robust Transformers using Robust Kernel Density Estimation**
XH, T. Ren, T. Nguyen, K. Nguyen, J. Ghosh, and N. Ho
Neural Information Processing Systems (NeurIPS) 2023
- [8] **Architecture Agnostic Federated Learning for Neural Networks**
D. Makhija, **XH**, N. Ho, and J. Ghosh
International Conference on Machine Learning (ICML) 2022
- [9] **Simultaneously Reconciled Quantile Forecasting of Hierarchically Related Time Series**
XH, S. Dasgupta, and J. Ghosh
Artificial Intelligence and Statistics (AISTATS) 2021
- [10] **Certified Monotonic Neural Networks**
Spotlight presentation: 280/9454 $\approx$ 2.96%; over 185 citations as of July 2026
X. Liu, **XH**, N. Zhang, and Q. Liu
Neural Information Processing Systems (NeurIPS) 2020

Other Papers

- [11] **MILM: Large Language Models for Multimodal Irregular Time Series with Informative Sampling**
H. Chung, S. Li, Y. Wald, **XH**, S. Saria, and J. Ghosh
Preprint 2026
- [12] **EvoMed-Agent: Self-Evolving Reasoners for Multimodal Clinical Sequences via Active Counterfactual Verification**
XH, Y. Wang, C. Chen, H. Chung, S. Li, G. Gudur, W. Dai, P. Liang, and S. Saria
In submission 2026
- [13] **MINT: Multimodal Instruction Tuning with Multimodal Interaction Grouping**
X. Shan*, Q. Cao*, **XH***, H. Yu*, and P. Liang
Preprint 2025
- [14] **Between Linear and Sinusoidal: Rethinking the Time Encoder in Dynamic Graph Learning**
H. Chung, S. Chaudhari, **XH**, Y. Wald, S. Saria, and J. Ghosh
Transactions on Machine Learning Research (TMLR) 2025
- [15] **SSL-DA: Semi- and Self-Supervised Learning with Dual Attention for Echocardiogram Segmentation**
L. Lv, **XH**, Z. Sun, Z. Li, X. Wang, T. Jiang, Y. Liu, T. Li, J. Xu, L. You, G. Yao, F. Sun, and J. Xing
Journal of Imaging Informatics in Medicine 2025
- [16] **Quadratic Gating Functions in Mixture of Experts: A Statistical Insight**

- P. Akbarian*, H. Nguyen*, **XH***, and N. Ho
In submission to SIAM Journal on Mathematics of Data Science 2024
- [17] **On Expert Estimation in Hierarchical Mixture of Experts: Beyond Softmax Gating Functions**
H. Nguyen*, **XH***, C. Harris, N. Ho, and S. Saria
In submission to JMLR 2024
- [18] **Achieving Fairness Across Local and Global Models in Federated Learning**
D. Makhija, **XH**, J. Ghosh, and Y. Kim
Preprint 2024
- [19] **Novel Node Category Detection Under Subpopulation Shift**
H. Chung, S. Chaudhari, Y. Wald, **XH**, and J. Ghosh
European Conference on Machine Learning and Data Mining (ECML-PKDD) 2024
- [20] **Efficient Forecasting of Large Scale Hierarchical Time Series via Multilevel Clustering**
XH, T. Ren, J. Hu, J. Ghosh, and N. Ho
International Conference on Time Series and Forecasting (ITISE) 2023
- [21] **A Novel Control-Variates Approach for Performative Gradient-Based Learners with Missing Data**
XH, J. Hu, and J. Ghosh
International Joint Conference on Neural Networks (IJCNN) 2023
- [22] **Dynamic Combination of Heterogeneous Models for Hierarchical Time Series**
XH, J. Hu, and J. Ghosh
International Conference on Data Mining Workshops (ICDMW) 2022
- [23] **Split Localized Conformal Prediction**
XH, Z. Tang, J. Ghosh, and Q. Liu
ICML Workshop on Distribution-Free Uncertainty Quantification 2021
- [24] **Multi-Pair Text Style Transfer for Unbalanced Data via Task-Adaptive Meta-Learning**
XH and J. Lundin
ACL MetaNLP Workshop 2021
- [25] **Model-Agnostic Explanations using Minimal Forcing Subsets**
XH and J. Ghosh
International Joint Conference on Neural Networks (IJCNN) 2021
- [26] **Transparent Interpretation with Knockout**
XH, Y. Feng, N. Zhang, and Q. Liu
ICML Workshop on Human Interpretability in Machine Learning (WHI) [Spotlight] 2020
- [27] **Sensing Personality to Predict Job Performance**
S. Lin, S.M. Mattingly, **XH**, P. Audia, et al.
CHI Workshop on The Future of Work 2019
- [28] **Effects of Integrated Intent Recognition and Communication on Human-Robot Collaboration**
A. Gutierrez, M.L. Chang, **XH**, and K.C. Chang
International Conference on Intelligent Robots and Systems (IROS) 2018

Patents

[1] Machine-Learning Based Generation of Text Style Variations for Digital Content Items

J. Lundin, O. Schoppe, **XH**, M. Sollami, B. Lonsdorf, A. Ross, D. Woodward, and S. Rohde

US Patent 2022/0245322 A1

2022

ACADEMIC EMPLOYMENTS

Postdoctoral Fellow, Johns Hopkins University

Baltimore, MD & Mountain View, CA

Supervisors: Prof. Suchi Saria, Prof. Gregory D. Hager

09/01/2023 – now

- Developing multi-modal Transformers and agents that integrate vital signs, lab results, clinical notes, and chest X-rays, enhancing predictive accuracy/reasoning of multimodal models with an emphasis on clinical applications

Graduate Research Assistant, The University of Texas at Austin

06/01/2019 – 08/14/2023

Intelligent Data Exploration and Analysis Lab, Department of Electrical and Computer Engineering

Supervisor: Prof. Joydeep Ghosh

Austin, TX

- Research topics include time-series forecasting, clustering, uncertainty quantification, federated learning, robustness and interpretability of machine learning

Data Science Group, Department of Statistics and Data Sciences

Supervisor: Prof. Nhat Ho

Austin, TX

- Research topics include optimal transport, Bayesian deep learning, model fusion/alignment, mixture of experts, and robustness of Transformers

Statistical Learning and AI Group, Department of Computer Science

Supervisor: Prof. Qiang Liu

Austin, TX

- Research topics include the robustness and interpretability of deep neural networks, conformal prediction, generative models, and predictive modeling of mobile sensor data

INDUSTRY EMPLOYMENTS

Google

Sunnyvale, CA

Research Scientist Intern (Host: Dr. Xu Gao)

05/31/2022 – 09/02/2022

- Enhanced the workload prediction accuracy for GCP machines by 20% compared to baseline
- Utilized explanation tools (Shapley value) to understand the root causes of machine downtime

Intuit

Mountain View, CA

Data Scientist Intern (Mentor: Dr. Jing Hu)

06/15/2021 – 09/03/2021

- Improved cash-flow forecasting accuracy by 10% and deployed models to applications
- Decreased computation time by 45% through the development of a parallelized program

Salesforce

San Francisco, CA

Research Intern (Mentor: Dr. Jessica Lundin)

06/01/2020 – 08/21/2020

- Implemented 3 different state-of-the-art language models (LM) for text style transfer
- Leveraged meta-learning to fine-tune pre-trained LM on a small and unbalanced dataset

CognitiveScale

Austin, TX

Applied Scientist Intern (Mentor: Dr. Suyog Jain)

05/29/2018 – 08/03/2018

- Reproduced and improved 5 different session-based recommendation algorithms
- Led a team project of 4 interns to research “Ethical AI” and pitched a business idea to company

GRANTS & PROTOCOLS

Grant Drafter and Key Personnel, Intuit AI Research, **\$300K** in total 2020 – 2023

- Drafted grant proposals and budget planning documents for ~ \$100,000/year from **Intuit AI Research** for three consecutive years. Developed a series of models in time-series forecasting and clustering, deployed in industry-level financial forecasting pipelines. Awarded annually to Dr. Joydeep Ghosh.

Core Contributor, Gordon and Betty Moore Foundation Grant, ~ **\$500K** in total 2023 – now

- Developed multimodal clinical foundation models and post-deployment continuous monitoring methods that served as core technical contributions. Awarded annually to Dr. Suchi Saria.

Protocol Drafter and Submitter, Johns Hopkins Medicine IRB 2024 – now

- Drafted, submitted, and maintained annual renewals for the following IRB protocols:
 - **IRB00464029**, “Leveraging Multimodal Patient Data for Enhanced Clinical Task Performance”
 - **CR00054024**, “Development and Validation of Robust Models for Adverse Event Risk Prediction”

COMMUNITY SERVICES

Reviewer, International Conference on Learning Representations (ICLR)	2020 – now
Reviewer, Neural Information Processing Systems (NeurIPS)	2020 – now
Reviewer, International Conference on Machine Learning (ICML)	2021 – now
Reviewer, Artificial Intelligence and Statistics (AISTATS)	2021 – now
Reviewer, International Joint Conference on Neural Networks (IJCNN)	2023 – now
Reviewer, Machine Learning for Health (ML4H)	2023 – now
Reviewer, International Conference on Acoustics, Speech and Signal Processing (ICASSP)	2023 – now
Reviewer, Machine Learning for Healthcare (MLHC)	2024 – now
Reviewer, SIAM Conference on Data Mining (SDM)	2026 – now
Journal Reviewer, Pattern Recognition	2019 – now
Journal Reviewer, Journal of Open Source Software	2024 – now
Journal Reviewer, IEEE Transactions on Pattern Analysis and Machine Intelligence (T-PAMI)	2025 – now
Journal Reviewer, Nature Biomedical Engineering	2025 – now
Journal Reviewer, Medical Image Analysis	2026 – now
Mentor, ML4H Author & Reviewer & Career Mentorship Program	2024 – now
Technical Program Committee, WellComp Workshop at UbiComp/ISWC	2026
Reading Group Organizer, Multimodal Learning in Health at JHU	2024 – 2025
Member, UT-Austin ECE Graduate Admission Committee	2021 – 2023
Volunteer, International Conference on Machine Learning (ICML) Workshops	2020, 2021, 2023
Third-year Class Representative, University of Edinburgh	2015 – 2016

SUPERVISION

Shravan Chaudhari, CS Ph.D. Student	JHU, 2023 –
Carl Harris, Biomedical Engineering Ph.D. Student	JHU, 2023 –
Chen Chen, CS M.Sc. Student	JHU, 2026 –
Yuxin Wang, Applied Mathematics M.Sc. Student	JHU, 2026 –
Jiarui Shao, CS M.Sc. Student	JHU, 2026 –
Tanvi Ranade, CS B.Sc. + M.Sc. Student	JHU, 2024 – 2026
Hsing-Huan Chung, ECE Ph.D. Student	UT-Austin, 2023 – 2026 (now with Uber)
Disha Makhija, ECE Ph.D. Student	UT-Austin, 2023 – 2024 (now with Amazon)

SELECTED HONOURS AND AWARDS

ICML Gold Reviewer Award	2026
NeurIPS Scholar Award, \$2000	2023 – 2024
Excellent Teaching Assistant Award, \$1000	2018
Outstanding Graduates of Beijing City (top 3%)	2017
University of Edinburgh 2 + 2 Scholarship, £2500 per year	2015 – 2016
Comprehensive Scholarship, ¥5000	2014 – 2015
China National Scholarship, ¥8000	2013 – 2014

TEACHING EXPERIENCE

Course Design and Trial Teaching, JHU Teaching Institute Certificate	2025
Teaching Assistant, CS 395T, Deep Probabilistic Modeling	UT-Austin, 2019
Teaching Assistant, ECE 422C, Software Design & Implementation	UT-Austin, 2018
Lab Instructor, ELEE09024, Digital System Design 3	University of Edinburgh, 2016

INVITED TALKS & PRESENTATIONS

Guiding Mixture-of-Experts in Multimodal Learning: Temporal Dynamics and Robust Fusion	<i>EcoSta</i> , 2025
FuseMoE: Mixture-of-Experts Transformers for Fleximodal Fusion	<i>MIT Media Lab</i> , 2025
FuseMoE: Mixture-of-Experts Transformers for Fleximodal Fusion	<i>Columbia University</i> , 2024
Reliable Predictions for Structured and Corrupted Data	<i>Fudan University</i> , 2023
Hierarchical Time Series Forecasting	<i>IIF Workshop on Forecast Reconciliation</i> , 2023
Reliable Prediction of Treatment Effect Through Time	<i>Intuit AI</i> , 2023
Time Series Reconciliation with Dynamically Changing Structures	<i>Google Cloud AI</i> , 2022
Mixture-of-Experts for Probabilistic Forecasts of Aggregated Time Series	<i>Intuit AI</i> , 2021
Certified Monotonic Neural Networks	<i>Salesforce Research</i> , 2020
Transparent Interpretation with Knockout	<i>Intuit AI</i> , 2019

TECHNICAL SKILLS

Programming: Python (expert); R, Matlab (proficient); Java, Verilog, C/C++ (knowledge of)
Frameworks: PyTorch, TensorFlow, Scikit-Learn, Wandb, HuggingFace, TIMM, NLTK
Language: English (fluent), Mandarin (native), French (beginner)
Others: $\LaTeX$, Shell Script, ROS, SQL

REFERENCES

Suchi Saria, John C. Malone Associate Professor (Postdoc Advisor)	ssaria@cs.jhu.edu
Department of Computer Science, Johns Hopkins University	Baltimore, MD & New York, NY
Joydeep Ghosh, Schlumberger Centennial Chair Professor (PhD Advisor)	jghosh@utexas.edu
Department of Electrical and Computer Engineering, UT-Austin	Austin, TX
Paul Pu Liang, Assistant Professor	ppli@mit.edu
Media Lab and EECS, MIT	Cambridge, MA